

AI-Powered UTI Antibiotic System

ANALYSIS REPORT • ACADEMIC RESEARCH DOCUMENT

Patient Demographics

Age / Gender	43.0 / Female
Department	Gynaecology
Diagnosis	Urethritis
UTI Classification	Recurrent UTI
Sample Type	Pus Swab

Clinical Presentation

Chief Complaints: Lower abdominal heaviness;Urinary burning

Comorbidities: None

Surgical History: None

Social History: Non smoker

Laboratory Results

Test Name	Value	Unit	Normal Range
Cbp Lymphocytes	38.0	%	20 - 40
Wbc	8920.0	cells/ μ L	4000 - 11000
Polymorphs	63.0	%	40 - 75
Crp	8.0	mg/L	< 10
Rft Serum Creatinine	0.7	mg/dL	0.6 - 1.3
Serum Uric Acid	4.0	mg/dL	3.5 - 7.2
Blood Urea	24.0	mg/dL	7 - 20
Cue Pus Cells	11.0	/HPF	0 - 5
Epithelial Cells	3.0	/HPF	0 - 5
Proteins	Trace		Negative
Rbc	1.0	/HPF	0 - 3

AI Pathogen Prediction

Predicted Organism	Enterococcus faecium
Bacteria Type	Gram-positive coccus (Higher risk of Vancomycin resistance - VRE)

Antibiotic Susceptibility Profile

Predicted Sensitive	Daptomycin
Predicted Resistant	Clarithromycin

Recommended Therapy

Daptomycin

Dosage: 6 mg/kg IV once daily (adjusted to 6 mg/kg for uncomplicated UTI)

Precautions: Normal renal function (serum creatinine 0.7 mg/dL); monitor creatinine and CPK; no known drug allergies; avoid in patients with myopathy or concurrent statin use; ensure adequate IV access; monitor for muscle pain

Rationale: Daptomycin is effective against Enterococcus faecium, including vancomycin-resistant strains, and is appropriate for recurrent UTI caused by VRE

Antibiotic Drug History

Clinical Summary

Infection Summary

- **Infection type:** Recurrent lower urinary tract infection (urethritis) in a 43-year-old female.
- **Causative organism:** Enterococcus faecium (gram-positive coccus, high risk of vancomycin resistance).
- **Antibiotic profile:**
- **Resistant:** Clarithromycin.
- **Sensitive:** Daptomycin (most reliable for VRE).

Recommended Therapy

- **Drug:** Daptomycin
- **Dosage:** 6 mg/kg IV once daily (adjusted to 6 mg/kg for uncomplicated UTI).
- **Rationale:** Daptomycin is bactericidal against E. faecium, including vancomycin-resistant strains, and is the preferred agent for recurrent UTI caused by VRE.

Precautions & Considerations

- Renal function is normal (creatinine 0.7 mg/dL); monitor serum creatinine and creatine phosphokinase (CPK) periodically.
- Avoid in patients with myopathy or concurrent statin therapy.
- Ensure adequate IV access and monitor for muscle pain or weakness.
- No known drug allergies reported.

This regimen targets the identified pathogen with a high likelihood of efficacy while mitigating potential adverse effects.